Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **secure cloud-native applications**, **microservices**, and **AI-enabled workflows** using React, TypeScript, Python, Java, SQL, Docker, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Architected **multi-tenant backend services** on Azure SQL/PostgreSQL and optimized query patterns, reducing dashboard latency from **1.2s to 350ms** while supporting 500+ production users.
- Built **FastAPI orchestration pipelines** for 5K+ monthly jobs with validation, reconciliation, and failure-isolation controls, improving release reliability and enabling scalable service operations.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **React + Flask full-stack applications** handling 10K+ daily grading requests with secure file ingestion, REST APIs, and concurrent processing for high-throughput education workflows.
- Implemented **CI/CD and automated quality controls** across model-backed services, reducing feedback turnaround by **60%** while maintaining stable production behavior under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **Python experimentation pipelines** over 100K+ trajectory logs with reproducible evaluation workflows, strengthening statistical analysis and model-debugging quality for AI system development.
- Delivered technical diagnostics and implementation summaries that accelerated translation of research findings into production-ready engineering plans and cross-team decisions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **API and ETL microservices** using Python/Java OOP, AWS Glue, and S3, improving response times by **30%** and supporting resilient data integration across distributed systems.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito in Agile sprints, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **MERN app modules** with REST APIs and WebSockets supporting eight workflows and 2M+ synchronized records, improving scalability and maintainability for client-facing operations.
- Automated **Docker/Jenkins deployment pipelines**, reducing deployment time by **50%** and improving release consistency for cross-functional engineering teams.

Skills

Languages: **Python**, **Java**, **JavaScript/TypeScript**, SQL, C++
Frontend: **React.js**, HTML5, CSS3, responsive UI development
Backend: **Flask**, **FastAPI**, Node.js, REST APIs, microservices
Cloud/DevOps: **AWS** (S3, Glue, RDS), Azure SQL, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD
Data/AI: PostgreSQL, MySQL, MongoDB, LLM-integrated workflows, model evaluation, production monitoring

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **scalable fintech platform** supporting 50K+ users and 1M+ transactions with secure API integrations, real-time synchronization, and Kubernetes-backed deployment reliability.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a **schema-aware SQL assistant** for 10+ databases with validated query generation, structured outputs, and cross-platform execution workflows for enterprise-style data operations.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Developed a **containerized RESTful service** with automated CI/CD releases, validation checks, and production deployment controls, achieving **87%** processing accuracy.